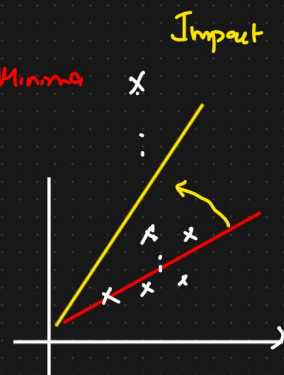
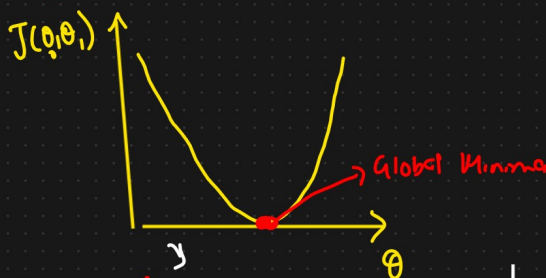
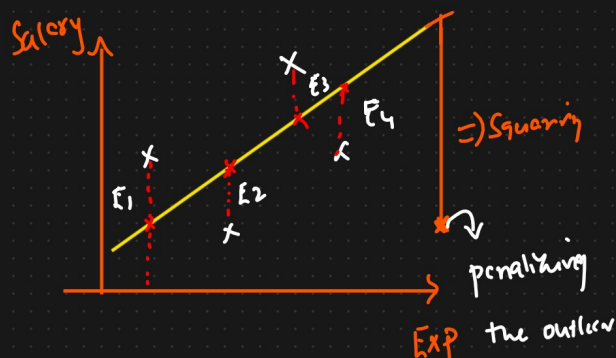


MSE, MAE, RMSE [Cost function] → Performance Metrics

R² and Adjusted R²



EXP	Salary
-	-
-	-
-	-

MSE ↑↑

MSE

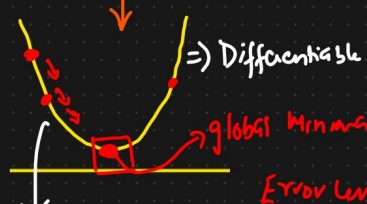
$$MSE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n} \Rightarrow \text{Cost function} \downarrow$$

Aim 1

Quadratic Equation

$$ax + by + c$$

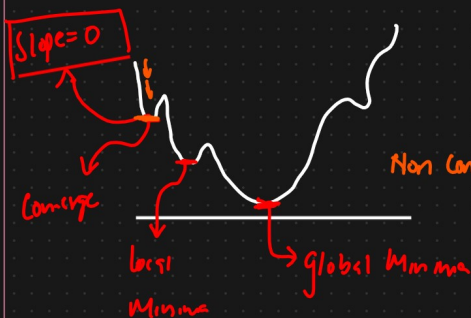
$$(a-b)^2 = a^2 - 2ab + b^2$$



Convex function

- Advantage
- ① Differentiable ✓
 - ② It has one local and one global Minima
 - ③ Converges faster

- Disadvantage
- ④ Not Robust to outliers
 - ⑤ It is not in same unit.



$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

EXP	Salary (laks)	$(y_i - \hat{y}_i)^2$ (lakh) ²
x	y	(MSE)

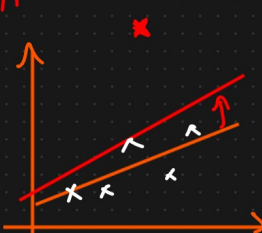
$$\text{Error } 2.5 \Rightarrow (lakh)^2 \Leftarrow$$

② Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

laks

MAE ↑



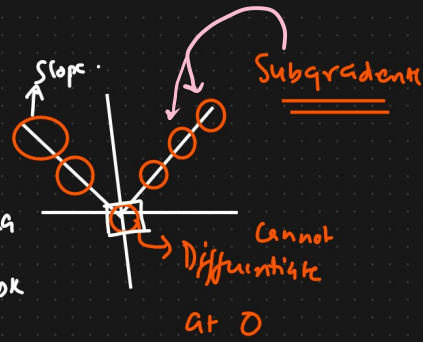
Compare to MSE in MAE the changes of line is less bcs we not squaring here.

Advantage

- ① Robust to outliers ✓
- ② It will be in the same unit

Disadvantage

- ④ Convergence usually take more time. Optimization is a complex task
- ⑤ Time consuming

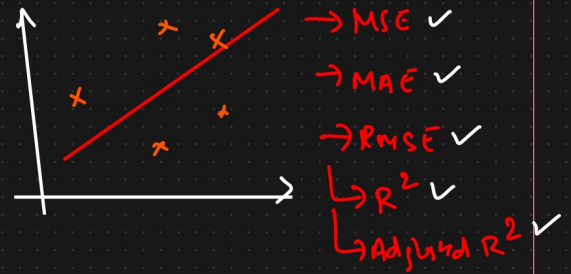
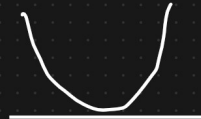


③ RMSE {Root Mean Squared Error}

MAE, MSE, RMSE

Performance metric ↑

$$\text{RMSE} = \sqrt{\text{MSE}}$$
$$= \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$



Advantage

- ① Same unit
- ② Differentiable

Disadvantage

- ④ Not Robust to outliers

MSE vs MAE vs RMSE

R^2 vs Adjusted R^2

When to use all of this?

